



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

- This report focuses on developing a predictive model to determine the success of Falcon 9 first-stage landings, a critical factor in SpaceX's cost-effective launch operations. With launch costs significantly lower than competitors (62 million USD vs. 165+ million USD), SpaceX's reusability advantage offers substantial savings. This analysis aims to empower alternative launch providers to compete effectively by understanding and predicting landing success, thus enabling optimized cost estimations and competitive bidding strategies.
- To achieve this, we undertook a comprehensive data science process. This involved meticulous data acquisition, rigorous data wrangling to ensure data quality, and insightful data visualization to uncover key patterns. We then developed and compared various machine learning models to accurately predict landing outcomes, focusing on identifying crucial factors for a safe launch, primarily using Exploratory data analysis (EDA).
- Our analysis revealed a significant correlation between **payload mass and landing success, particularly highlighting that successful drone ship landings are most likely to occur with payloads between 4000 and 6000 kg**. This finding, validated through rigorous model comparison and cross-validation, and using a dataset of past Falcon 9 launches, provides actionable insights for optimizing launch parameters and improving landing success rates which we will share in the detail reports. Our best performing model, a Decision Tree achieved an accuracy of 90%".
- This finding, verified by performing various model's to ensure the predictive accuracy, provides actionable insights for optimizing launch parameters and improving landing success rates which we will share in the detail reports.
- Ultimately, this report delivers a powerful tool for businesses seeking to challenge SpaceX's dominance in the commercial space launch market. By leveraging our predictive model and insights, companies can make data-driven decisions regarding launch costs, risk assessments, and competitive strategies, ultimately fostering a more competitive and innovative space launch landscape

Introduction

- How SpaceX is so success? Could we learn something from SpaceX experience and perform better for effective and safe journey to the outerspace?
- We found the Analytics Models that shown 90% probability we could send people to the space and back safely based on our learning on success and failed space launch over the decade.

Section 1

Methodology

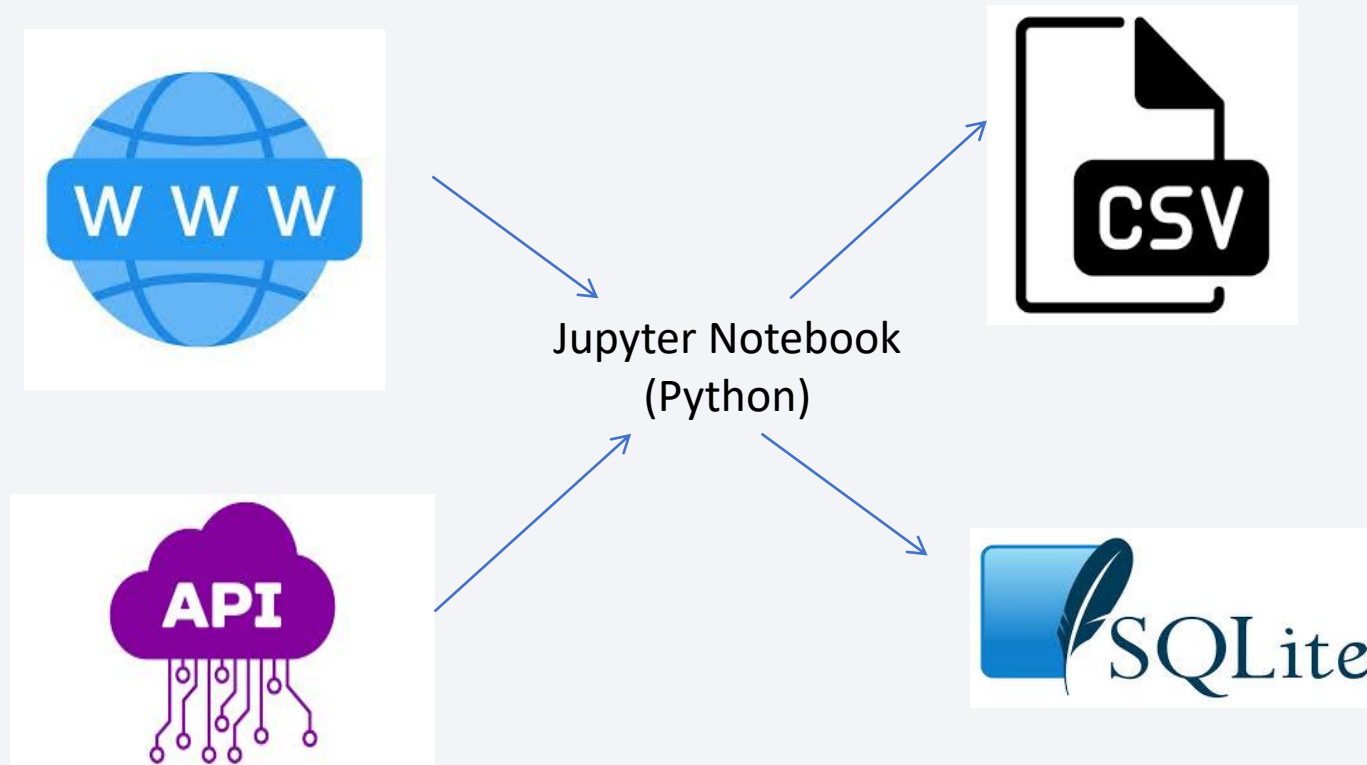
Methodology

Executive Summary

- Data collection methodology:
 - Using Data Scraping and also combine with data collection through API.
- Perform data wrangling
 - To ensure the data are able to be visualized and provide meaningful primary understanding on the data.
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
 - We various predictive models and use train and test data to verify the models and find the best models in term of accuracy/

Data Collection

- Describe how data sets were collected.



The illustration show how we collect data by performing:

1. Data Web Scraping

2. Get Data from API

3. Merge the result as dataframe in Python

4. Save the data in CSV format

5. Also save the data in SQLite

These data is used for EDA and build the Predictive Models.

Data Collection – SpaceX API

- Present your data collection with SpaceX REST calls using key phrases and flowcharts
- Add the GitHub URL of the completed SpaceX API calls notebook (must include completed code cell and outcome cell), as an external reference and peer-review purpose

Using SpaceX data from below API:

<https://api.spacexdata.com/v4/>



Jupyter Notebook
(Python code, see Github
link below)



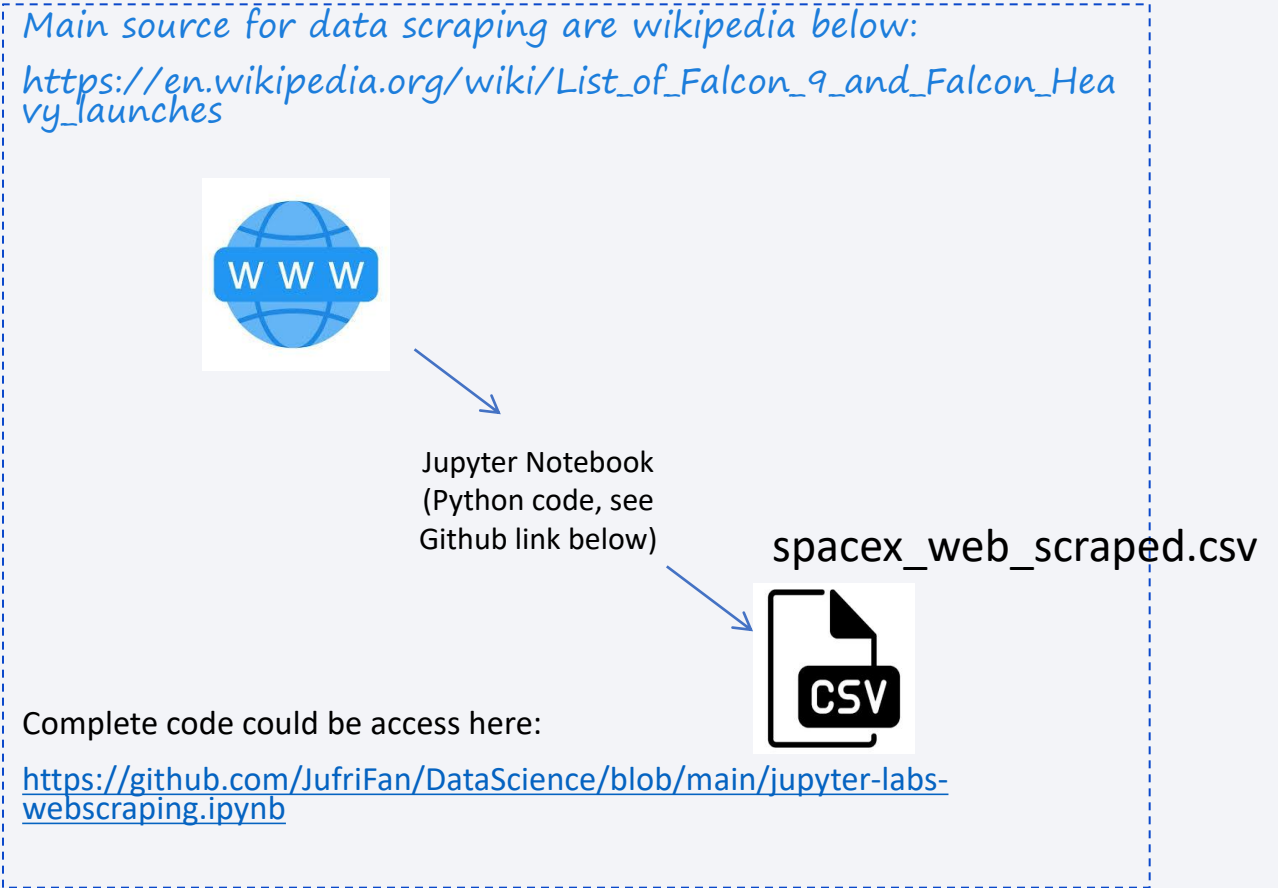
dataset_part_1.csv

Complete code could be access here:

<https://github.com/JufriFan/DataScience/blob/main/jupyter-labs-spacex-data-collection-api.ipynb>

Data Collection - Scraping

- Present your web scraping process using key phrases and flowcharts
- Add the GitHub URL of the completed web scraping notebook, as an external reference and peer-review purpose



Data Wrangling

- After collecting the data, I check the data in term of completeness (null), data type, etc. and I notice some data is not complete, null thus data wrangling needed.
- Since bad outcomes are based on landed on various condition (land, ocean or ground pad). I need to simplify it into 1 columns.
- I also added Class to identify success or failed landing.
- After I finished the data wrangling, I save the dataframe into a CSV file (dataset_part_2.csv)
- Complete code is in this link:
<https://github.com/JufriFan/DataScience/blob/main/labs-jupyter-spacex-Data%20wrangling.ipynb>

EDA with Data Visualization

- To make sense of the data, I am using these charts and plotted it to perform Exploratory Data Analysis:
 - Visualizing Categorical Data (using catplot) using LaunchSite, PayloadMass and FlightNumber
 - Visualizing using barplot to find relationship between success rate and orbit type.
 - Visualized using scatterplot to find any relationship between FlightNumber and Orbit type or relationship between Payload Mass and Orbit type
 - Lastly I am using lineplot to show average success rate, to get the average launch success trend from 2010 to 2020
 - Summarize that payload mass is significant factors to be explore futher and success rate is increase over the decades which is a good sign.
- The GitHub URL
(<https://github.com/JufriFan/DataScience/blob/main/edadataviz.ipynb>)
explain the completed EDA with data visualization notebook

EDA with SQL

- Using SQL, I perform validation on data collected such as:
 - `SELECT distinct(Launch_Site) FROM SPACEXTABLE;`
 - `SELECT Launch_Site FROM SPACEXTABLE WHERE Launch_Site LIKE 'CCA%' LIMIT 5;`
 - `SELECT Booster_Version, PAYLOAD_MASS_KG_, Landing_Outcome FROM SPACEXTABLE WHERE Landing_Outcome = 'Success' AND PAYLOAD_MASS_KG_ BETWEEN 4000 AND 6000;`
- The GitHub URL (https://github.com/JufriFan/DataScience/blob/main/jupyter-labs-eda-sql-coursera_sqlite.ipynb) explain the completed EDA with SQL notebook

Build an Interactive Map with Folium

- I am using Folium to show the LaunchSite as map objects such as markers, circles, lines, to verify launch site success and failure with market and circle. Lines as used to calculate the distance of interest objects to the launch site.
- Added those objects enable a visualized of launch site and in case failure, I can assure no casualties around the launch site.
- The GitHub URL (https://github.com/JufriFan/DataScience/blob/main/lab_jupyter_launch_site_location.ipynb) explain the completed interactive map with Folium map

Build a Dashboard with Plotly Dash

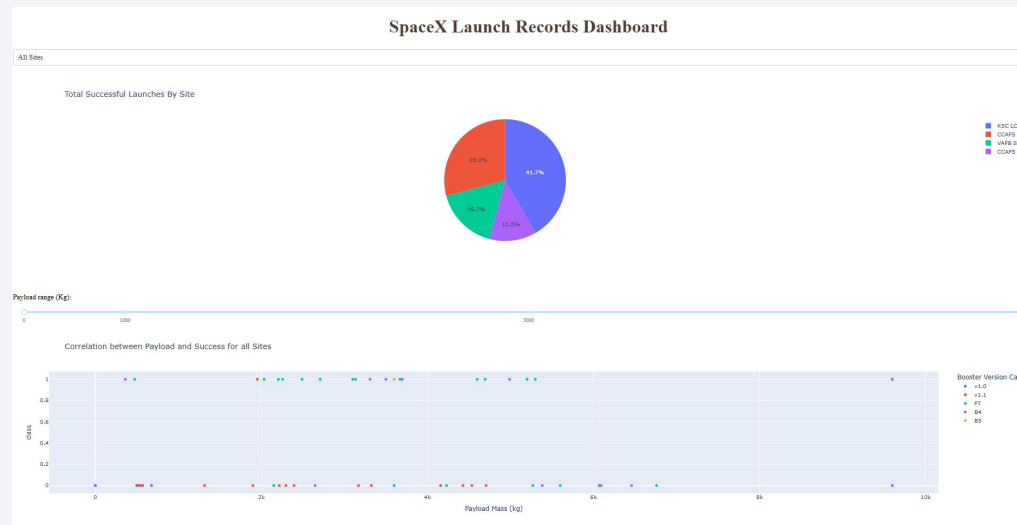
- I created an interactive dashboard to show plots/graphs and based on the launch site and payload. This dashboard allow users to play around with the data and show the data in realtime.
- I added payload and launchsite as this data is important and imapcted to success or failure of launch and landing.
- The GitHub URL (https://github.com/JufriFan/DataScience/blob/main/spacex_dash_app.py) explain the completed Plotly Dash lab
- As this is a python code, you need to deploy it to jupyter notebook.

Predictive Analysis (Classification)

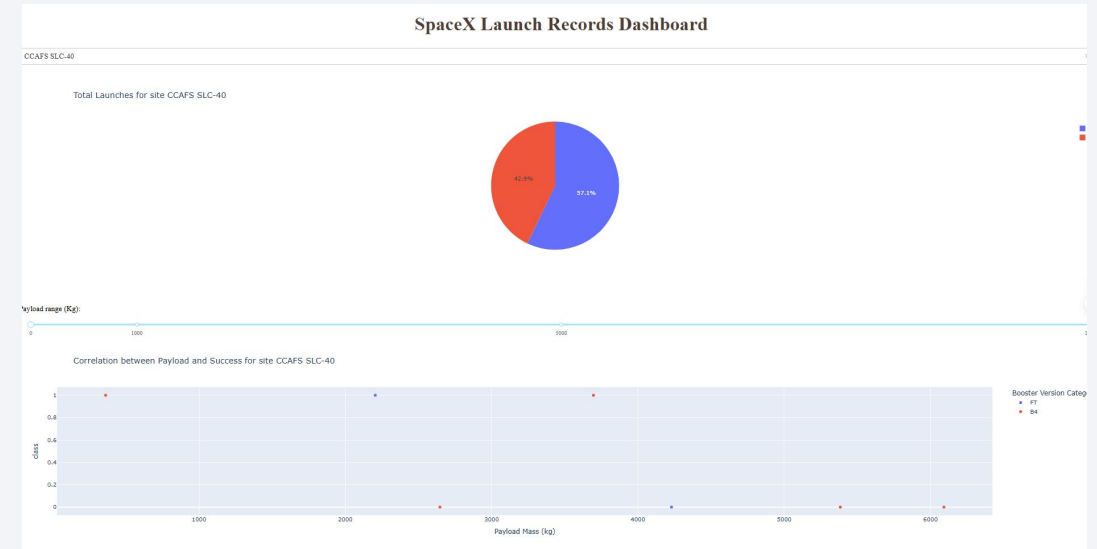
- This is what I built, evaluated, improved, and found the best performing classification model
 - Model LogisticRegression on train data perform 86.25% but on test data only 70%.
 - An SVM model, or Support Vector Machine model on training data perform 85% accuracy and even better to 90% on test data.
 - Decision tree model on training data perform 90% accuracy and also even better to 95% on test data.
 - KNN model perform 87.85% on training data but perform only 75% on test data.
- I recommend using Decision Tree model for consistency result.
- The GitHub URL (https://github.com/JufriFan/DataScience/blob/main/SpaceX_Machine%20Learning%20Prediction_Part_5.ipynb) explain the completed predictive analysis lab

Results

- Interactive analytics demo in screenshots



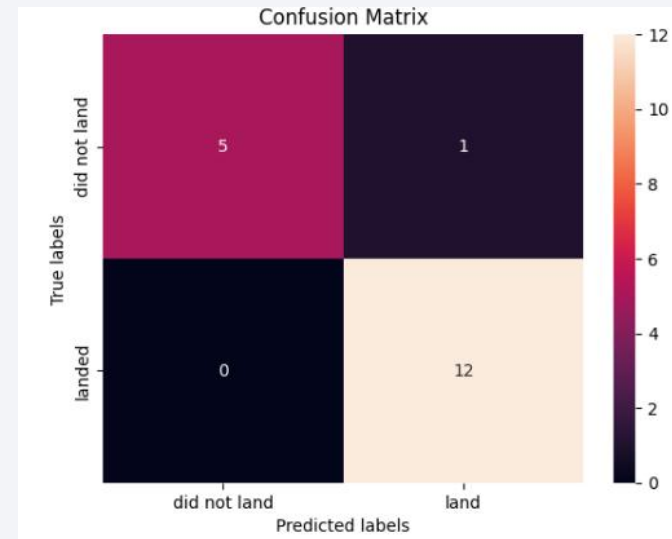
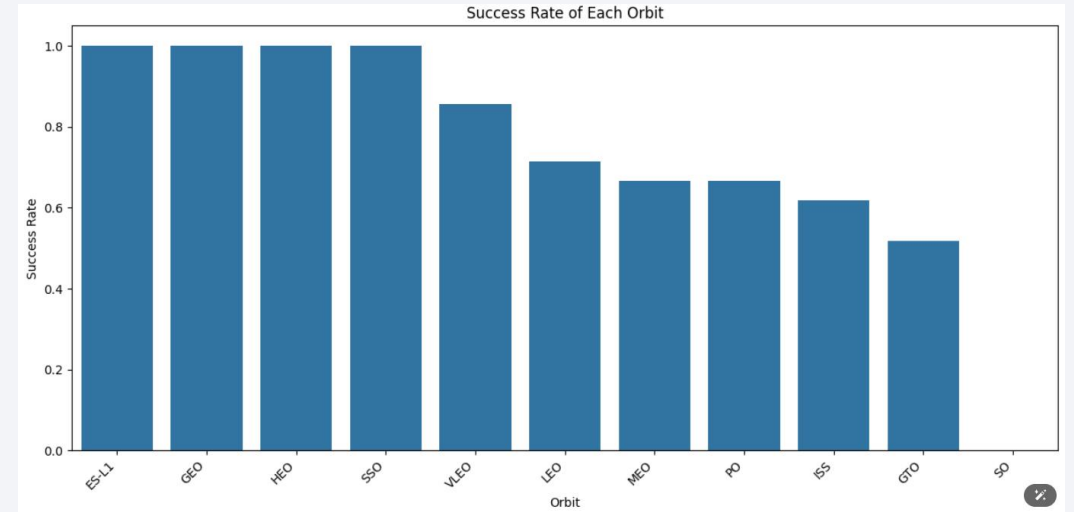
All Launch sites

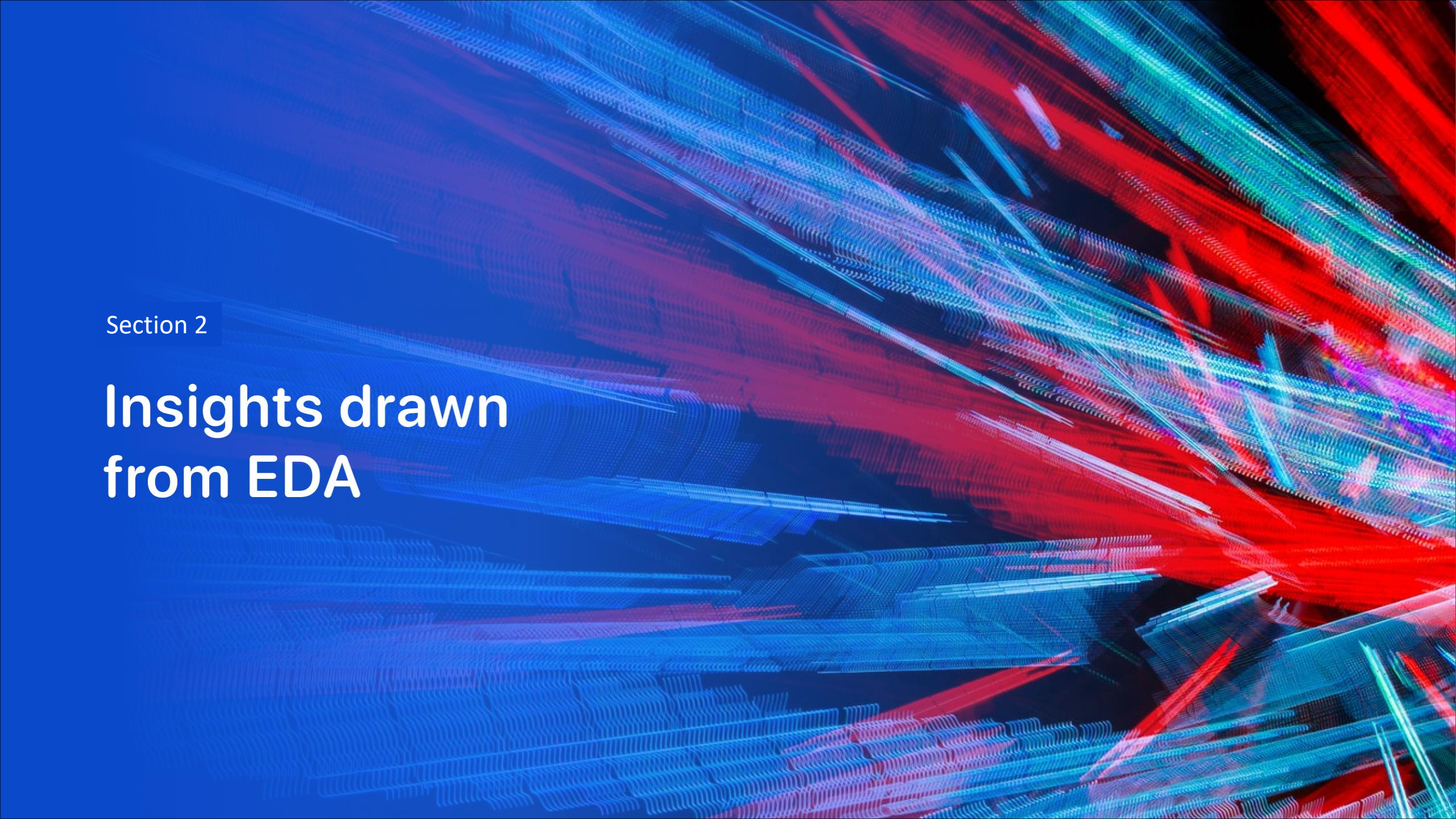


Select specific Launch sites

Results

- Exploratory data analysis results (landing site with highest success result)
- Predictive analysis results (Decision tree models)



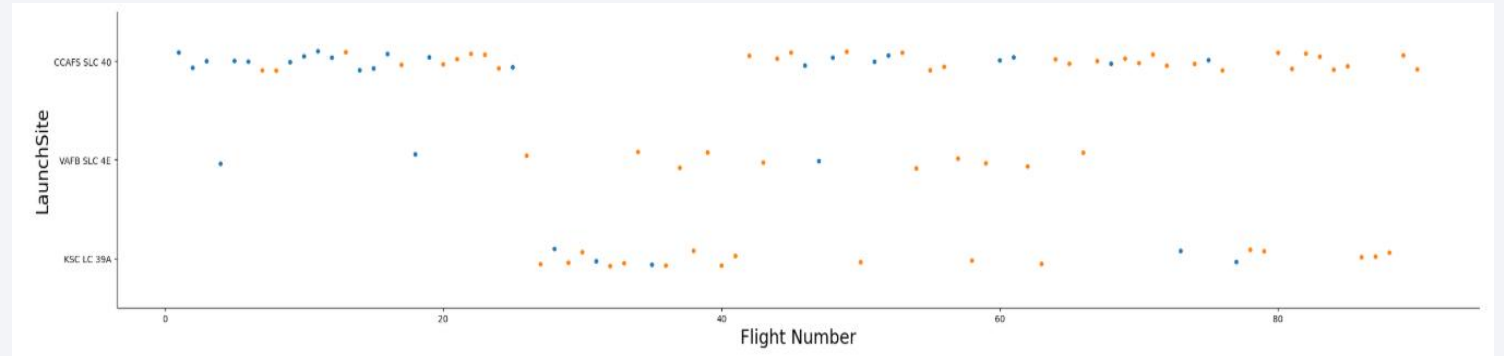
The background of the slide is an abstract composition. It features a dark blue base color. Overlaid on this are numerous diagonal streaks in shades of red and cyan. A faint, light blue grid pattern is also visible, particularly in the lower-left quadrant. The overall effect is dynamic and technological.

Section 2

Insights drawn from EDA

Flight Number vs. Launch Site

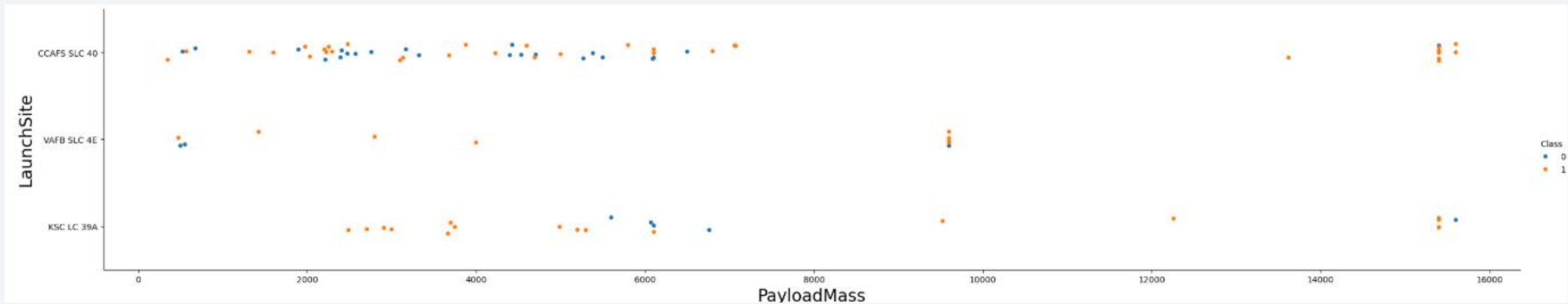
- Show a scatter plot of Flight Number vs. Launch Site



- CCAFS-SLC 40 has highest launch and success launch and landing

Payload vs. Launch Site

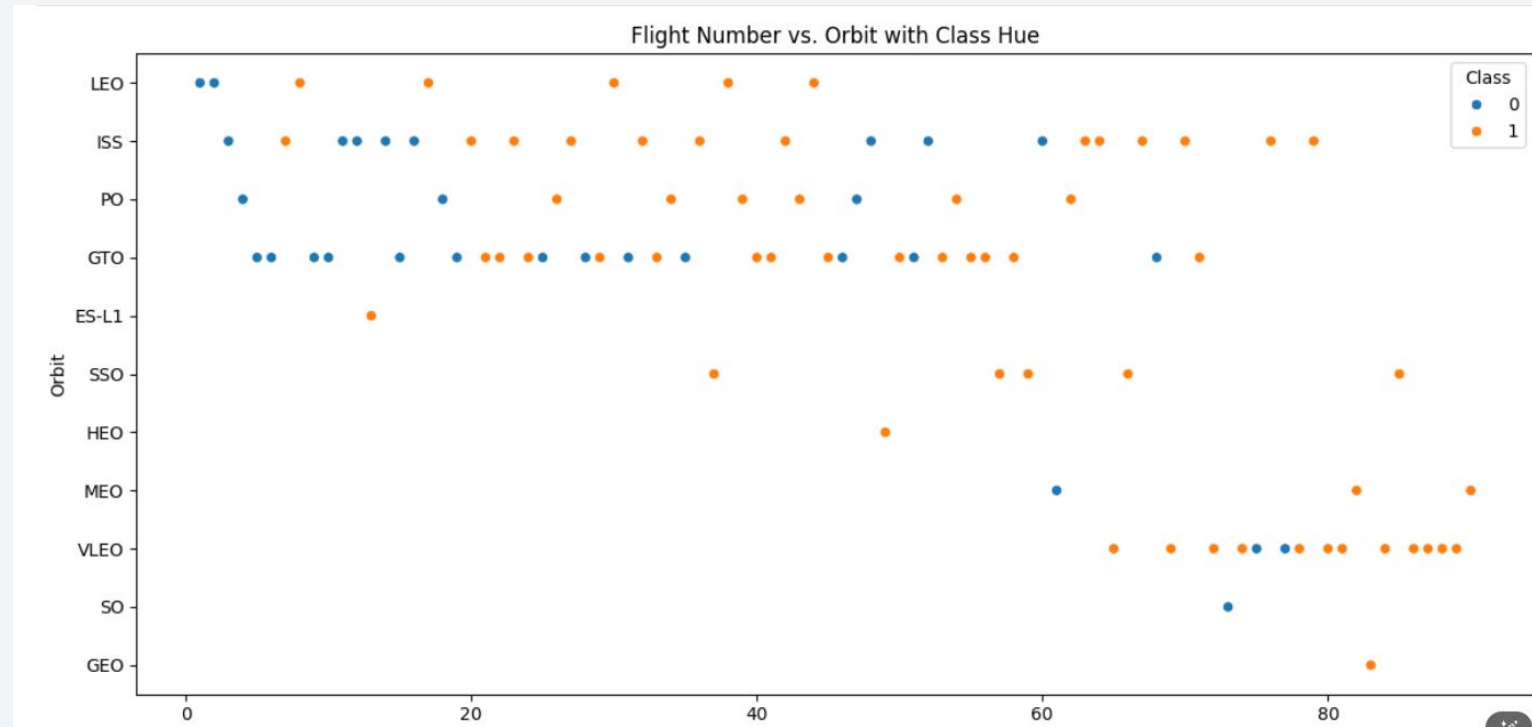
- Show a scatter plot of Payload vs. Launch Site



- Payload under 10000KG have a greater success landing.

Success Rate vs. Orbit Type

- Show a bar chart for the success rate of each orbit type
- the more flight, all shown all orbit type has more success landing. thus orbit type may not have relation with success or failure of a flight



Flight Number vs. Orbit Type

- Show a scatter point of Flight number vs. Orbit type
- Show the screenshot of the scatter plot with explanations

Payload vs. Orbit Type

- Show a scatter point of payload vs. orbit type
- Show the screenshot of the scatter plot with explanations

Launch Success Yearly Trend

- Show a line chart of yearly average success rate
- Show the screenshot of the scatter plot with explanations

All Launch Site Names

- Find the names of the unique launch sites
- Present your query result with a short explanation here

Launch Site Names Begin with 'CCA'

- Find 5 records where launch sites begin with 'CCA'
- Present your query result with a short explanation here

Total Payload Mass

- Calculate the total payload carried by boosters from NASA
- Present your query result with a short explanation here

Average Payload Mass by F9 v1.1

- Calculate the average payload mass carried by booster version F9 v1.1
- Present your query result with a short explanation here

First Successful Ground Landing Date

- Find the dates of the first successful landing outcome on ground pad
- Present your query result with a short explanation here

Successful Drone Ship Landing with Payload between 4000 and 6000

- List the names of boosters which have successfully landed on drone ship and had payload mass greater than 4000 but less than 6000
- Present your query result with a short explanation here

Total Number of Successful and Failure Mission Outcomes

- Calculate the total number of successful and failure mission outcomes
- Present your query result with a short explanation here

Boosters Carried Maximum Payload

- List the names of the booster which have carried the maximum payload mass
- Present your query result with a short explanation here

2015 Launch Records

- List the failed landing_outcomes in drone ship, their booster versions, and launch site names for in year 2015
- Present your query result with a short explanation here

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order
- Present your query result with a short explanation here

A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The image is a deep blue, with the horizon line visible. The city lights are concentrated in the lower right quadrant, showing a dense network of urban areas. The text "Section 3" is overlaid on the left side of the image.

Section 3

Launch Sites Proximities Analysis

<Folium Map Screenshot 1>

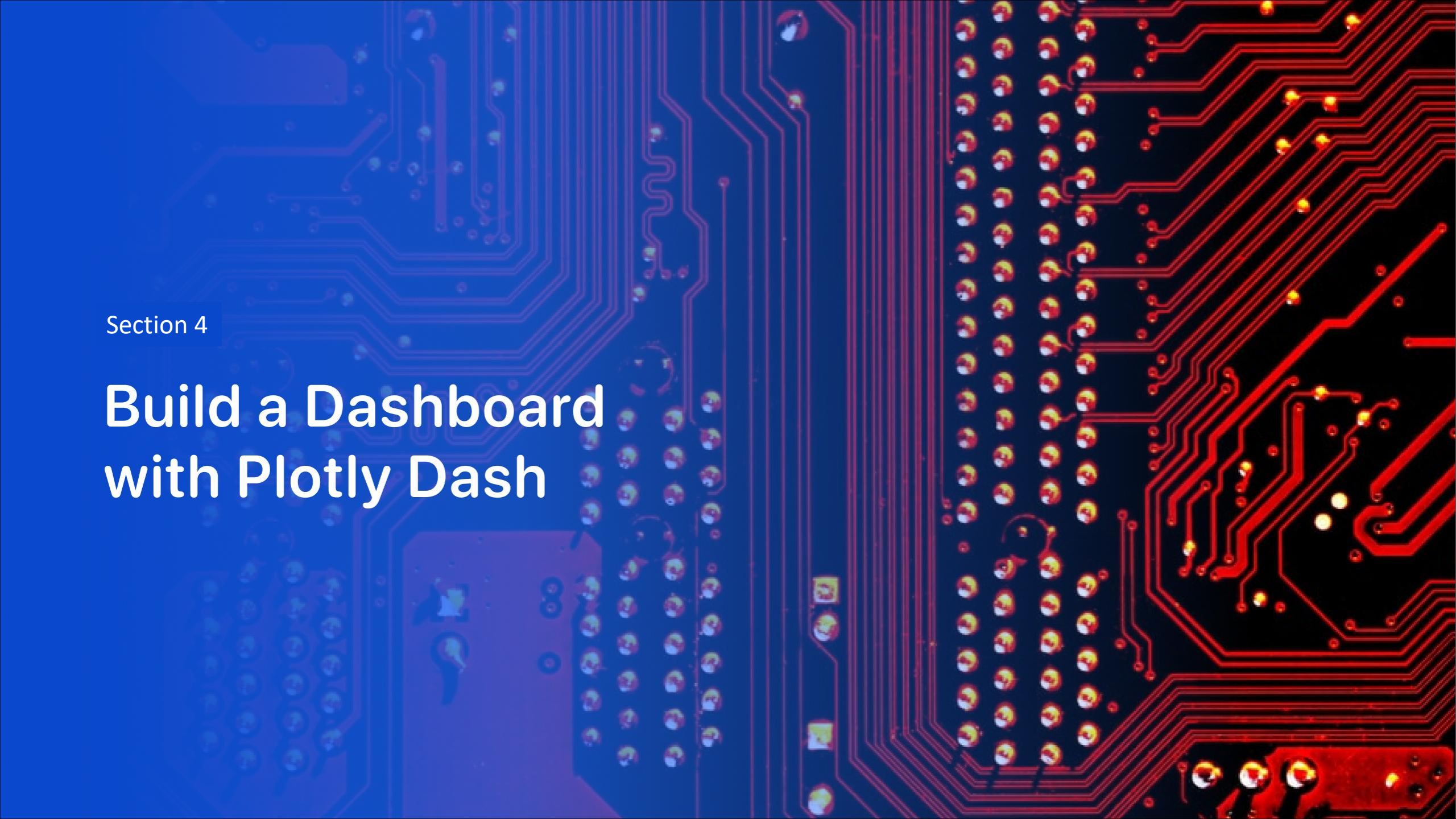
- Replace <Folium map screenshot 1> title with an appropriate title
- Explore the generated folium map and make a proper screenshot to include all launch sites' location markers on a global map
- Explain the important elements and findings on the screenshot

<Folium Map Screenshot 2>

- Replace <Folium map screenshot 2> title with an appropriate title
- Explore the folium map and make a proper screenshot to show the color-labeled launch outcomes on the map
- Explain the important elements and findings on the screenshot

<Folium Map Screenshot 3>

- Replace <Folium map screenshot 3> title with an appropriate title
- Explore the generated folium map and show the screenshot of a selected launch site to its proximities such as railway, highway, coastline, with distance calculated and displayed
- Explain the important elements and findings on the screenshot



Section 4

Build a Dashboard with Plotly Dash

<Dashboard Screenshot 1>

- Replace <Dashboard screenshot 1> title with an appropriate title
- Show the screenshot of launch success count for all sites, in a piechart
- Explain the important elements and findings on the screenshot

<Dashboard Screenshot 2>

- Replace <Dashboard screenshot 2> title with an appropriate title
- Show the screenshot of the piechart for the launch site with highest launch success ratio
- Explain the important elements and findings on the screenshot

<Dashboard Screenshot 3>

- Replace <Dashboard screenshot 3> title with an appropriate title
- Show screenshots of Payload vs. Launch Outcome scatter plot for all sites, with different payload selected in the range slider
- Explain the important elements and findings on the screenshot, such as which payload range or booster version have the largest success rate, etc.



Section 5

Predictive Analysis (Classification)

Classification Accuracy

- Visualize the built model accuracy for all built classification models, in a bar chart
- Find which model has the highest classification accuracy

Confusion Matrix

- Show the confusion matrix of the best performing model with an explanation

Conclusions

- Point 1
- Point 2
- Point 3
- Point 4
- ...

Appendix

- Include any relevant assets like Python code snippets, SQL queries, charts, Notebook outputs, or data sets that you may have created during this project

Thank you!

